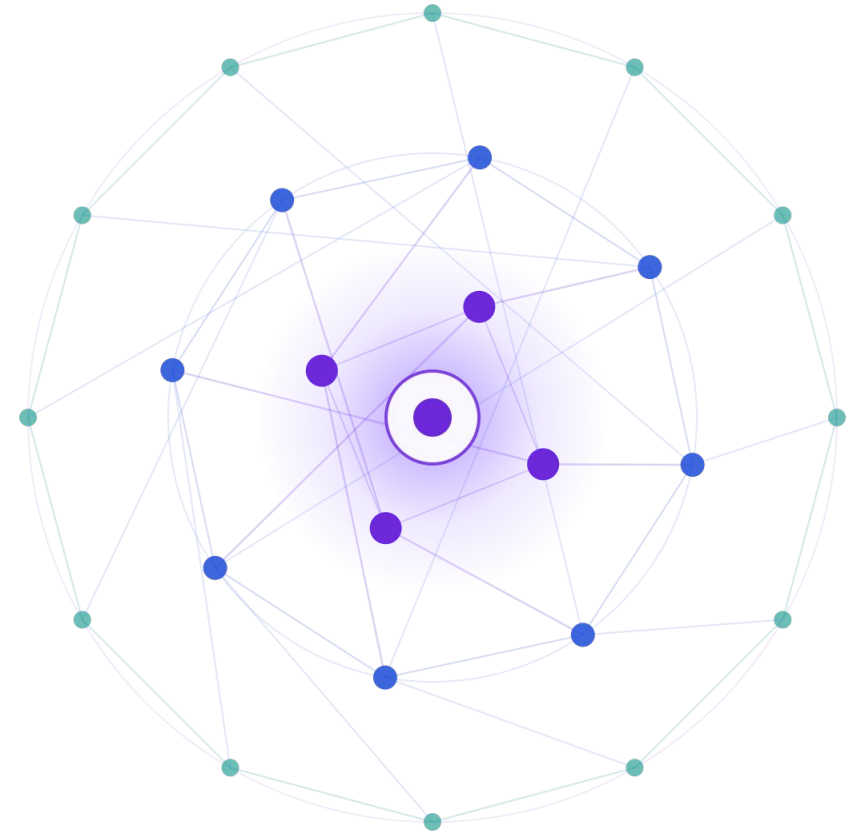


Generative AI Hackathon

3rd Edition 2026 – 27

- Theme ID - 05
- Team Name - BILIO
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- Submission Github link - <https://github.com/rishithraswant-eng/bilio>



Theme

Theme 05: Interruptible Real-Time Agents

Voice-native assistance with fast-and-slow execution and robust interruption handling.

The problem

Users interrupt, correct details and change plans while an assistant is still working.

Why it matters

Stale tool calls, lost slots and duplicate actions can turn a correction into a wrong booking or an incorrect dispatch.

BILIO's approach

Acknowledge immediately, cancel stale work and re-plan using the corrected session state.

Existing Solutions & Gaps

Existing approaches referenced in the submission

Google Duplex, Amazon Alexa, Apple Siri, OpenAI Voice Mode and sequential LLM agents.

Execution and cancellation gaps

Blocking workflows; stale results after corrections; no explicit `call_id` cancellation contract.

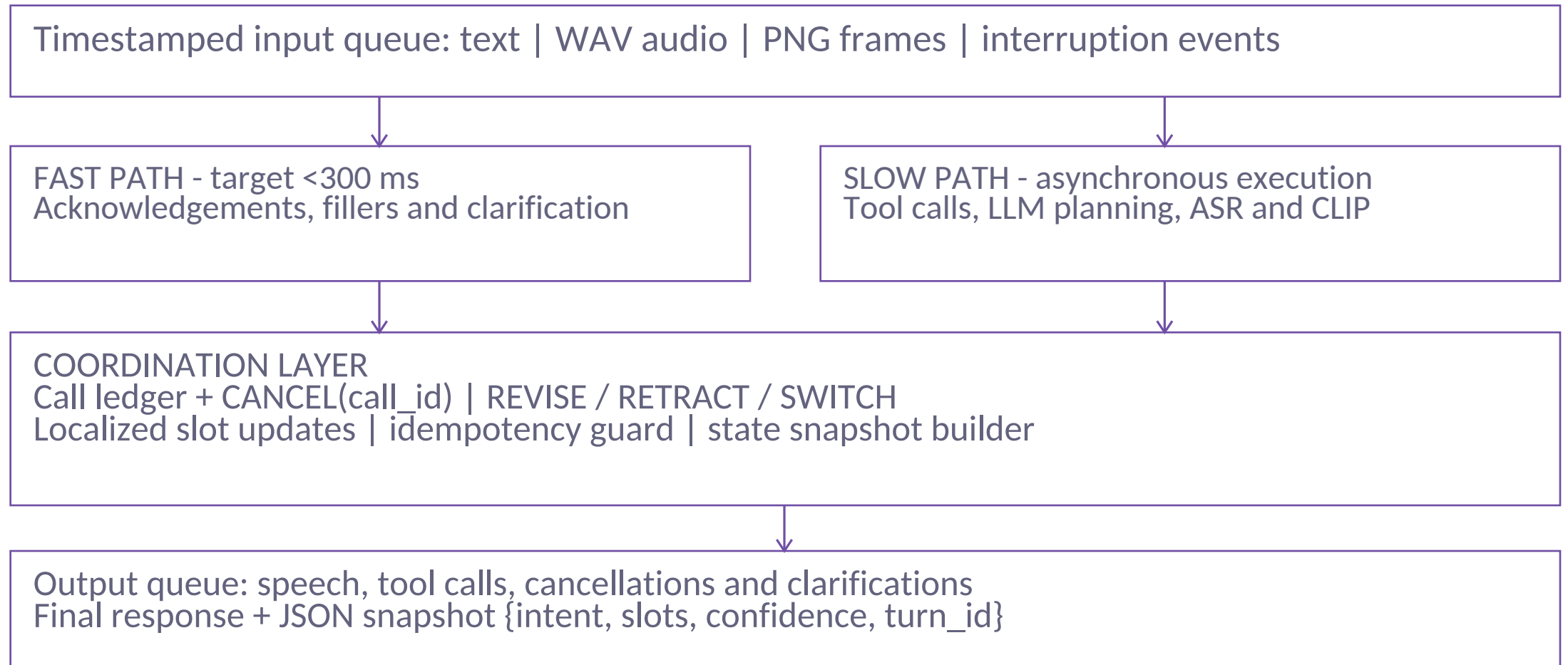
State and action-safety gaps

Session slots can be lost; duplicated state-changing actions need an operation ledger.

Grounding and verification gaps

Multimodal confidence gating, schema-driven tools and structured state snapshots are the capabilities BILIO is designed to bring together.

Our Solutions & Architecture Diagram



Demo & Product Walkthrough

1. Flight correction: Mumbai → Pune

Cancel call_id_001; acknowledge the change; search Pune with tomorrow and economy preserved.

2. Full retraction: “Actually never mind”

Cancel call_id_002; create no support ticket; emit a retracted intent with cleared slots.

3. Audio grounding: “What’s that sound?”

Acknowledge the WAV clip; run ASR + audio classification; offer a relevant follow-up search.

4. Line: NH48 → ECR, two people injured

Abort the proposed dispatch; correct the location; request confirmation before finalizing.

Tools and tech stack used

Runtime and real-time transport

Python 3.12, asyncio, LiveKit and LiveKit Agents SDK.

Speech and visual grounding

Deepgram Nova-3 / TTS, Silero VAD, NeMo ASR fallback, Gemini; CLIP + confidence gate.

LLM planning and fallbacks

Gemini 2.0 Flash → Cerebras Llama → OpenRouter. GPT-4o used only as evaluation judge.

Evaluation and infrastructure

Full-Duplex-Bench v3, virtual clock replay, mock tools, jsonschema; FastAPI, Docker / docker-compose and GitHub.

Impact & Use case

In-car navigation and customer support

Change a destination mid-route request or a seat preference mid-booking; suppress stale calls.

Field and consumer troubleshooting

Ground answers in camera frames; ask for clarification when visual confidence is low.

Accessibility

Handle hesitations and self-repairs while preserving the user's intent and unchanged slots.

Emergency triage

Require explicit confirmation before dispatch; abort pending actions after location corrections.

Innovation highlights, results and limitations

Innovation

Dual execution paths, call-specific cancellation, localized slot updates, schema-driven tool parsing and structured state snapshots.

Results reported in the submission

29/30 held-out scenarios (96.7%); 229/229 tests passing.
Offline text replay: 61/100 baseline; live judged score pending.

Known limitations

In-memory simulated tools; single replica; regex-based location extraction; no geocoding.
Live LiveKit evaluation awaits the final network run.

Runtime constraints reported as met

Python 3.12; 120 s wall-clock cap; 300 s warmup; session-only memory; no cross-session cache.

What's next

Complete live evaluation

Run the pending LiveKit network evaluation and report judged scores alongside the baseline.

Move beyond simulated tools

Proposed next step: connect real booking / dispatch APIs while retaining confirmation and idempotency.

Improve location handling and scale

Proposed next step: add geocoding and replace the single-replica runtime with a scalable deployment.

Finalize submission materials

Add the actual team repository, member details, reproducible README and a demo video under 5 minutes.

Brownie points slide(differentiation)

Triage Line: confirmation before consequential action

PROPOSED → PENDING_CONFIRMATION → FINALIZED.

Corrections before finalization abort the pending dispatch and trigger re-planning.

Safety and verifiability built into execution

Operation ledger suppresses duplicate state changes; each final response includes a JSON snapshot.

Confidence-aware multimodal behavior

CLIP confidence below 0.55 triggers clarification instead of an unsupported visual answer.

Checklist- Updated on Public GitHub

- Working prototype code — public or shared GitHub repo (Y/N)
- README with reproducible setup instructions (Y/N)
- Demo video, max 5 minutes (YouTube or Drive link)
- Presentation file (PPT or PDF) (Y/N)

Submission status (in the same order as the checklist)

1. Repository: final public / shared URL to be added.
2. README: reproducible setup instructions to be confirmed.
3. Demo video: YouTube / Drive link to be added.
4. Presentation: Yes - this PDF; GitHub upload to be confirmed.

Thank you

Organised by the Language AI Team and the PRISM Team, Samsung R&D Institute India